

XAI: Generalized Additive Models & Prototype-Based Approaches

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

Paper Presentations ---

Reminder

If you are presenting next week, please come and see us in office hours this week. Ideally, we would like to walk through your slides with you.

- Each paper: **25 mins** of Paper Presentation + **5 mins** Q&A



Guidelines for Paper Presentation ---

- 1 Motivation
- 2 Problem Statement
- 3 Summary of Contributions
- 4 Related Work
- 5 Preliminaries + Background (*Intuition First!*)
- 6 Approach (*Intuition First!*)
- 7 Key Experimental Results
- 8 Conclusions
- 9 Your Perspective on the Weaknesses of Paper
- 10 What would you do differently?

Generalized Additive Models

🍷 Intelligible Models for HealthCare ---



Intelligible Models for HealthCare

Caruana et al.

Contributions ---

Main Contributions

Two case studies where Generalized Additive Models (intelligible) yield state-of-the-art accuracy:

- Pneumonia risk prediction
- 30-day hospital readmission

Claim

GAMs is a class of models that can handle the **interpretability/accuracy trade-off** quite well.

Roadmap ---

- ➊ Motivation
- ➋ Intelligible Models
- ➌ Case study: Pneumonia risk
- ➍ Case study: 30-day readmission

🍷 Motivation (1/3) ---

- A large project to evaluate application of ML to healthcare problems
- Predicting **probability of death (POD)** for pneumonia patients

Most Accurate

Neural nets: **0.86 AUC**

Actually Deployed

Logistic Regression: **0.77 AUC**

Why was logistic regression used instead?

Motivation (2/3) ---

- Rule-based learning method was also used
- Insight: $\text{HasAsthma}(x) \Rightarrow \text{LowerRisk}(x)$
 - ▶ *Counterintuitive?*

Key Problem

Rule-based system was intelligible, making it easy to **recognize and remove dangerous rules**. Lack of intelligibility made it harder to deploy neural nets because it was difficult to know other problems with the model.

🍷 Motivation (3/3) ---

- Many more models today are equally unintelligible: SVMs, random forests, boosted trees

Why GAMs?

- GAMs are both **intelligible and accurate!**
- **Editable** by domain experts

🍷 GAMs -- Shape Functions (Bike Sharing) ---



🍷 GAMs -- Shape Functions (Concrete/Other) ---



GAMs and GA²Ms -- Formulas ---

GAM

$$g(E[y]) = \beta_0 + \sum_j f_j(x_j)$$

GA²M (with pairwise interactions)

$$g(E[y]) = \beta_0 + \sum_j f_j(x_j) + \sum_{i \neq j} f_{ij}(x_i, x_j)$$

- g : link function — identity (regression) or $\log(E[y]/1 - E[y])$ (classification)
- f_j : shape function

🍷 Intelligibility and Accuracy ---



Shape Functions -- Learning ---

- **Regression Splines**
- **Trees**
- **Ensembles of Trees**

Learning GA²Ms

- 1 Represent each component as a spline or regression tree on a single/pair of features
- 2 Gradient boosting with bagging of shallow trees
- 3 Build GAM first, then detect and rank all possible pairs of interactions in the residual
- 4 Choose top k pairs (determined by CV)

Case Study: Pneumonia Risk ---

Dataset

- 14,199 pneumonia patients
- Train: 9,847
- Test: 4,352
- 46 features

Task

Predict **probability of death (POD)**

10.86% of patients died (1,542)

🍷 Pneumonia Risk: Features ---



🍷 Pneumonia Risk: AUC Results ---



Result

GAM (0.8542) and GA²M (0.8576) **outperform Random Forests (0.8460) and LogitBoost (0.8493)**, while remaining intelligible.

🍷 Understanding Outputs of GA^2M (1/2) ---



🍷 Understanding Outputs of GA²M (2/2) ---



Insight

Childhood cancers are associated with **high risk of death** from pneumonia.

Case Study: 30-Day Readmission ---

Dataset

- 195K patients (train)
- 100K patients (test)
- **3,956 features**

Context

Hospitals with high readmission rates are **penalized financially** (inadequate earlier care).

8.91% of patients readmitted within 30 days.

🍷 30-Day Readmission: AUC Results ---



Result

GA²M (0.7833) matches LogitBoost (0.7835) and outperforms Random Forests (0.7671), while remaining fully interpretable.

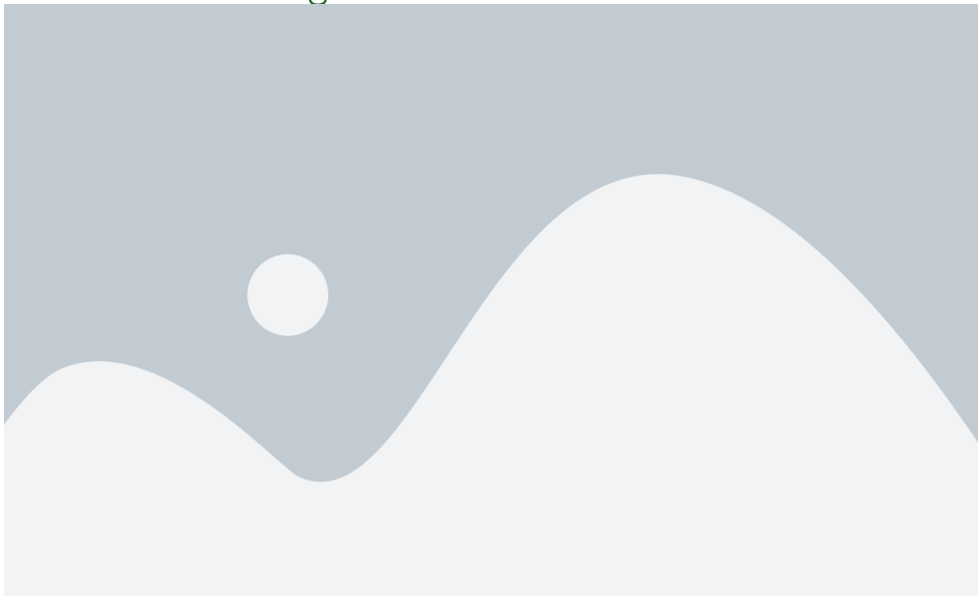
🍷 Patient Level Insights (1/2) ---



High Risk Patient — $p(\text{risk}) = 0.9326$

Lots of admissions · Received lot of Amoxicillin (strep/pneumonia) · Verapamil (hypertension)

🍷 Patient Level Insights (2/2) ---



🍷 Modularity of GAMs ---

Model Decomposition

$$\text{prediction} = \underbrace{\beta_0}_{\text{bias}} + \underbrace{\sum_j f_j(x_j)}_{\text{individual features}} + \underbrace{\sum_{i \neq j} f_{ij}(x_i, x_j)}_{\text{pairwise interactions}}$$

This structure helps us **clearly understand the model**: for each patient, we can compute which term is resulting in what risk score and rank terms based on their contribution.

🍷 Feature Shaping vs. Expert Discretization ---

Expert Discretization

Experts provide inputs by manually discretizing features, used in logistic regression.

GAM Feature Shaping

GAMs **learn** the function shapes automatically — and outperformed expert-discretized LR.

Conclusion

Feature shaping is **valuable** — GAMs captured non-linearities that experts missed.

🍷 Correlation $\neq$ Causation ---

⚠ Important Warning

GAMs and GA²Ms are intelligible — **but they are not causal!**

- What we see in plots are **associations** captured from data, not causal implications
- It is often easy to confuse intelligibility of predictive models with causality

Please don't make that mistake!

Prototype-Based Approaches

🍷 Deep Learning for Case-Based Reasoning through Prototypes ---



Deep Learning for Case-Based Reasoning through Prototypes

Oscar Li, Hao Liu, Chaofan Chen, Cynthia Rudin

Contributions (Prototypes) ---

Novel Architecture

Proposed and developed a novel network architecture for deep learning that:

- Explains its own reasoning for each prediction
- **Not** post-hoc explanations
- Prototypes are learned **during training**
- Explanations are **faithful** to what the network computes

Motivation (Prototypes) ---

- ML models are increasingly deployed to answer societal questions $\Rightarrow$ **interpretability/transparency**
- **Radiology:** lack of transparency poses challenges to FDA approval for deep learning models
- Neural nets are particularly difficult to understand because of the high degree of non-linearity

🍲 Related Work: Post-hoc Explanations ---

Problem with Post-hoc Approach

Past neural nets were designed mainly for accuracy, with post-hoc explanations added after.

- Build neural net first, then interpret!
- Post-hoc explanations **may not be faithful** to the model
- Easy to create **multiple conflicting yet convincing** explanations, none of which is correct

🍷 Background: Autoencoder (1/3) ---



Non-linear Dimensionality Reduction and Reconstruction

🍷 Background: Autoencoder (2/3) ---



Background: Constructing an Autoencoder ---

- Constrain the number of nodes present in the hidden layer(s) of the network
 - ▶ Limiting the amount of information that can flow through the network
- By penalizing the network according to the **reconstruction error**, the model can learn the most important attributes of the input data

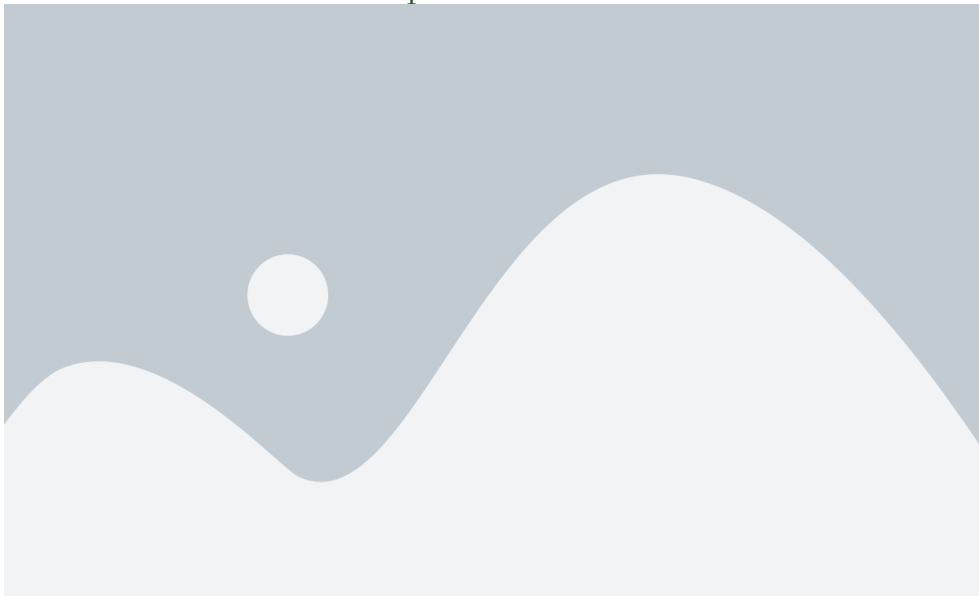
Key Insight

The encoding will learn and describe **latent attributes** of the input data.

🍷 Proposed Network Architecture ---



🍷 Autoencoder -- Latent Space ---



🍷 Prototype Layer ---



🍷 Fully Connected Layer ---



🍷 Softmax Layer ---



🍲 Advantages of the Proposed Architecture ---

Key Advantages

- **Automatically learns** useful features (non-linear dimensional reduction)
- Suitable for **high-dimensional data** such as images
- Prototype vectors can be **decoded and visualized** (same latent space as encoded inputs)
- Ability to interpret **without post-hoc analysis**

Cost Function ---

Cross-Entropy Loss

$$E(h \circ f, D) = \frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K -\mathbb{I}[y_i = k] \log((h \circ f)_k(\mathbf{x}_i))$$

Reconstruction Error

$$R(g \circ f, D) = \frac{1}{n} \sum_{i=1}^n \|(g \circ f)(\mathbf{x}_i) - \mathbf{x}_i\|_2^2$$

Cost Function: Interpretability Regularizers ---

R_1 : Prototypes close to training data

$$R_1(\mathbf{p}_1, \dots, \mathbf{p}_m, D) = \frac{1}{m} \sum_{j=1}^m \min_{i \in [1, n]} \|\mathbf{p}_j - f(\mathbf{x}_i)\|_2^2$$

R_2 : Training data close to prototypes

$$R_2(\mathbf{p}_1, \dots, \mathbf{p}_m, D) = \frac{1}{n} \sum_{i=1}^n \min_{j \in [1, m]} \|f(\mathbf{x}_i) - \mathbf{p}_j\|_2^2$$

- R_1 : Each prototype vector should be as close as possible to **at least one training example**
- R_2 : Each training example should be as close as possible to **one prototype**

Full Cost Function ---

$$L((f, g, h), D) = E(h \circ f, D) + \lambda R(g \circ f, D) + \lambda_1 R_1(\mathbf{p}_1, \dots, \mathbf{p}_m, D) + \lambda_2 R_2(\mathbf{p}_1, \dots, \mathbf{p}_m, D)$$

Training: Neural Network Steps ---

- 1 Define architecture
- 2 Outline cost function
- 3 Forward pass, compute derivatives, backpropagate, update parameters — repeat!

Note on min functions

Min functions are not technically differentiable — but in practice, packages allow it. This is essentially **gradient descent**.

🍷 Backpropagation: Intuition (1/4) ---



🍷 Backpropagation: Intuition (2/4) ---



🍷 Backpropagation: Intuition (3/4) ---



🍷 Backpropagation: Intuition (4/4) ---



Local gradient $\times$ upstream gradient

🍷 Results: MNIST Data ---



🍷 Learned Weight Matrix ---



🍲 Ablation Study on Cars Data (1/2) ---



🍲 Ablation Study on Cars Data (2/2) ---



Thank You!

References --- |